

## Assignment - 5.

### (1) FPGA vs CPLD

| Feature         | FPGA                          | CPLD                         |
|-----------------|-------------------------------|------------------------------|
| Logic           | Large number of LUTs, FF, RAM | Similar number of macrocells |
| Routing         | Complex                       | Simpler                      |
| Program ability | usually SRAM based            | EEPROM based                 |
| Speed           | High                          | Fast                         |
| Application     | CPU, DSP                      | control logic state machine. |

(2). Disjoint vs universal vs wilton

| Feature     | Disjoint | universal | wilton                    |
|-------------|----------|-----------|---------------------------|
| Routing     | low      | high      | very high                 |
| Switches    | fewer    | more      | moderate                  |
| Area        | low      | high      | moderate                  |
| Power       | low      | high      | moderate                  |
| congestion  | higher   | lower     | moderate                  |
| Performance | Good     | Good      | lowest<br>generally best. |

### 3) FPGA programmable interconnection

| Type     | reprogrammability | configuration       | security         | non volatile | Application   |
|----------|-------------------|---------------------|------------------|--------------|---------------|
| SRAM     | Excellent         | External needed     | low              | No           | HP<br>FPGA    |
| Antifuse | One time          | permanent           | high             | Yes          | Aero<br>space |
| EEPROM   | Good              | Retained internally | better than SRAM | Yes          | Embedded      |

(4) Xilinx Virtex-II FPGA Architecture

CLB: main logic element

Slices: each CLB containing 2 slices

LUTs: 4-input LUTs

FF: used for storage

Block RAM: dedicated memory blocks

DCMs: provide clock multiplication

I/O Block: Interface FPGA logic

Programmable: connects CLB, RAM, I/O, routing, etc.